← Back **Life Through Time**
Graded Assignment • 30 min

Due Mar 18, 11:59 PM CET

1. Which of these are typical adaptations seen in thermophiles? 1 point

- [ ] tolerance of osmotic stress
- [x] modified cell membrane composition
- [ ] cell dormancy
- [x] Thermostable proteins and enzymes

2. An organism which grows at which temperature is considered to be 'psychrophilic'? 1 point

- ( ) 50-80 degrees Celsius
- (•) Less than 15 degrees Celsius
- ( ) 30 degrees Celsius
- ( ) 110 degrees Celsius

3. Which of these extremes would you expect to encounter in space? 1 point

- [x] Low gravity
- [x] high levels of radiation
- [x] lack of oxygen
- [x] freezing temperatures

4. What name is given to organisms which can tolerate multiple extremes at once? 1 point

- ( ) Halophile
- ( ) Piezophile
- ( ) Xerophile
- (•) Polyextremophile

5. What can we gain from studying life in extreme environments? 1 point

- [ ] Proof of life on Mars
- [x] Understanding of the emergence of life on Earth
- [x] Understanding of the boundaries of life in the Universe
- [ ] Proof that life on Earth originated on another planet and was transferred here

6. What is the upper temperature limit of life? 1 point

- ( ) 110 degrees Celsius
- ( ) 204 degrees celsius
- ( ) 80 degrees Celsius
- (•) 121 degrees Celsius

7. In rocks of what age do we find fossils of the Ediacaran Fauna? 1 point

- (•) 585-542 Million Years Old
- ( ) 200 Million years old
- ( ) 2.5-3 billion years old
- ( ) 540 Million years Old

8. Which of these are advantages of multicellularity? 1 point

- [x] development of complex behaviour
- [x] increased specialization
- [x] increased size
- [x] physical protection

9. How many big mass extinction events have been found in the fossil record? 1 point

- ( ) 6
- ( ) 10
- ( ) 1
- (•) 5

10. What usually happens to biodiversity after a big extinction event? 1 point

- ( ) biodiversity never recovers
- (•) biodiversity recovers or increases
- ( ) biodiversity always reaches previous levels but never increases above the previous level

11. What innovation is represented by the Cambrian Explosion? 1 point

- [ ] first land plants
- [x] evolution of hard body parts such as shells.
- [ ] first multicellular organisms
- [ ] dinosaurs became extinct

12. By what mechanisms did multicellularity possibly arise? 1 point

- [x] cell clustering
- [x] incomplete division
- [ ] evolving bigger cells
- [ ] dividing really quickly

13. Three billion years ago, what percentage of the present level of oxygen was in the atmosphere? 1 point

- ( ) 0%
- ( ) more than 1%
- (•) less than 0.1%
- ( ) 21%

14. Select from the list below both the name and the timing of the sharp rise in oxygen in the Earth's atmosphere? 1 point

- [ ] Great Oxidation Biodiversification Event
- [x] 2.4 billion years ago
- [ ] 3.5 billion years ago
- [x] Great Oxidation Event

15. Which of these minerals cannot be used to infer the concentration of oxygen in the atmosphere at the time of formation? 1 point

- ( ) Pyrite
- ( ) Siderite
- ( ) Uraninite
- (•) Quartz

16. What is a palaeosol? 1 point

- (•) A fossilised soil
- ( ) A type of mineral
- ( ) A rock which forms under the sea
- ( ) A portable shelter on a hot day

17. The depletion of which element in palaeosols older than 2.4 billion years old is seen as evidence that the oxygen concentration in the atmosphere had changed over time? 1 point

- ( ) Pyrite
- ( ) Oxygen
- (•) Iron
- ( ) Banded Iron Formations

18. What effect did the Great Oxidation Event have on the climate? 1 point

- ( ) It caused the climate to become hotter
- (•) It caused the climate to cool

19. What benefits did the increase in oxygen have for life on Earth? 1 point

- [ ] There was a mass extinction because of the rise of oxygen
- [x] Organisms could become larger
- [x] It helped in the evolution of complex life
- [ ] The increased oxygen was toxic to some organisms

20. What disadvantage did the rise in oxygen have for life on Earth? 1 point

- (•) Oxygen was toxic to some organisms.
- ( ) The oceans became too warm for some organisms to survive.
- ( ) Organisms could become larger.
- ( ) Enough energy could be gained through oxygen respiration to allow complex life to evolve.